Industrial Safety Processes

Safety by Design

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Content

- The Concept of Safety by Design
- Theoretical Foundations
- Application
- Procurement

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Australian Study (2004)

"The main finding from the study is that design continues to be a significant contributor to work-related serious injury in Australia. This is the case with a wide variety of machinery, plant and equipment, although the extent of involvement varies between them."

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AIHS (Australian Institute of Health and Safety). (2019). Health and Safety in Design. In *The Core Body of Knowledge for Generalist OHS Professionals*. Tullamarine, VIC: Australian Institute of Health and Safety, p. 4.

Australian Study (2004)

"Most of the main design problems are old issues, with guarding the most prominent example. Other identified problems were poorly situated controls; inadequate interlock safety systems; absent or inadequate rollover protective structures and/or associated seat belts; inadequate fall protection; failed hydraulic lifting systems; and inadequate protection mechanisms (such as enclosed cabins) on mobile plant and vehicles."

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AIHS (Australian Institute of Health and Safety). (2019). Health and Safety in Design. In *The Core Body of Knowledge for Generalist OHS Professionals*. Tullamarine, VIC: Australian Institute of Health and Safety, p. 4.

Australian Study (2004)

"Design issues are rarely considered comprehensively in OHS research…The analysis has shown that:

- similar design problems are involved in many fatal accidents
- design is an important contributor to fatal injury in many industries, and
- solutions already exist for most of the identified design problems"

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AIHS (Australian Institute of Health and Safety). (2019). Health and Safety in Design. In *The Core Body of Knowledge for Generalist OHS Professionals*. Tullamarine, VIC: Australian Institute of Health and Safety, pp. 4-5.

Australia (2014)

"around 36% of in scope workplace deaths over the period 2006 to 2011 were definitely or possibly attributable to unsafe design of machinery, plant and powered tools.."

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AIHS (Australian Institute of Health and Safety). (2019). Health and Safety in Design. In *The Core Body of Knowledge for Generalist OHS Professionals*. Tullamarine, VIC: Australian Institute of Health and Safety, p. 5.

United Kingdom (2005 Paper)

"Research in the United Kingdom (UK) indicates that approximately 64% of injuries sustained in the construction industry are attributable to poor design in one way or another…[F]ailure to design and implement physical safeguards caused 35% of fatalities and more than 20% of non-fatal maintenance incidents in the UK petrochemical industry…"

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AIHS (Australian Institute of Health and Safety). (2019). Health and Safety in Design. In *The Core Body of Knowledge for Generalist OHS Professionals*. Tullamarine, VIC: Australian Institute of Health and Safety, p. 5.

Safety by Design

"Safety through design is defined as integrating hazard analysis and risk assessment methods early in the design and redesign processes and taking the actions necessary so that the risks of injury or damage are at an acceptable level."

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Hagan, Philip E., Montgomery, John F., & O'Reilly, James T., eds. (2015). *Accident Prevention Manual for Business & Industry: Engineering & Technology* (14th Edition). Itasca, Ill.: National Safety Council., p. 5.

Safety by Design

"…design can be considered a staged, iterative process that involves multiple stakeholders and must take account of the complexity, flexibility and dynamic nature of sociotechnical systems where the human is the key element in the system."

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AIHS (Australian Institute of Health and Safety). (2019). Health and Safety in Design. In *The Core Body of Knowledge for Generalist OHS Professionals*. Tullamarine, VIC: Australian Institute of Health and Safety, p. 2

Safety by Design

- It is the approach that applies safety control principles in advance to eliminate or effectively control hazards which are known or anticipated in an operation, process, or facility
- This is the proactive safety approach that – if done right – will significantly increase the level of effective safety

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Benefits

- Avoiding incidents and all the costs associated with them – suffering, insurance premiums, lost time, morale, reputation, replacement costs, etc.
- More efficient operational outcomes, usually linked to increased productivity and quality
- Establishes a strong physical climate of safety that will impact worker and management mindsets and activity

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The Approach

- Safety by design can be incorporated at all stages of a facility, process, or operational life cycle
- However, the greatest “bang for your buck” is realized when safety is considered and built into systems at the conceptual and design stages
- Revising systems after they are in place can be difficult, time consuming, disruptive, and more expensive – potentially more costly than anticipated incident costs

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Prevention

- The main theoretical view centers on system design that avoids or reduces the requirement for the worker decision-making
- Human factors engineering is the modern development of this approach, but is often limited to simple worker-system interface on a ergonomic or hygiene level
- Safety by Design is concerned with the entire process, facility, or operation

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Theory – Alphonse Chapanis

"The improvement in system performance that can be realized from the redesign of equipment is usually greater than the gains that can be realized from the selection and training of personnel."

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Hagan, Philip E., Montgomery, John F., & O'Reilly, James T., eds. (2015). *Accident Prevention Manual for Business & Industry: Engineering & Technology* (14th Edition). Itasca, Ill.: National Safety Council., p. 13.

Chapanis Importance

- Notice that Chapanis talks about improvement in system performance, not safety performance
- The key insight is that it is cheaper and more effective to design the system properly – to meet the requirements of the work and the worker – than to build a system that you then need to carefully select or repeatedly train workers on how to use

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Chapanis Expanded

- On a micro-scale, this is in line it human factors
- On a macro-scale, it points to the design of facilities, operations, and processes to function as a coherent enterprise in their own right and with linked operations

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Theory – William Haddon

- Focus on unwanted or harmful transfer of energy
- Approach is to ensure systems control possible releases of "energy" that could cause incidents
- Limited in application due to difficult focus on "energy"
  - Chemical exposure, psychological stressors, biological agents

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Hagan, Philip E., Montgomery, John F., & O'Reilly, James T., eds. (2015). *Accident Prevention Manual for Business & Industry: Engineering & Technology* (14th Edition). Itasca, Ill.: National Safety Council., pp. 13-14.

Prevention Through Design (PtD)

2007: the US National Institute for Occupational Safety and Health (NIOSH) began the Prevention through Design (PtD) initiative

2011: the American National Standards Institute published ANSI/ASSE Z590.3 *Prevention through design: Guidelines for addressing occupational hazards and risks in design and redesign processes*.

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AIHS (Australian Institute of Health and Safety). (2019). Health and Safety in Design. In *The Core Body of Knowledge for Generalist OHS Professionals*. Tullamarine, VIC: Australian Institute of Health and Safety, p. 9.

Safety by Design – Hierarchy of Controls

- Safety by Design is primarily concerned with Eliminating hazards in the workplace or identifying Engineering controls to mitigate the potential of those hazards to cause harm
- These are often the most difficult types of hazard controls to implement, and may be the most costly in terms of up-front costs
- See Hazard Assessment presentation later in this course

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Methods of Application

- Hazard and Operability Study (HAZOP)
- Failure Modes & Effects Analysis (FMEA)
- Cognitive Work Analysis (CWA)
- PETE Analysis
- Misconception Analysis

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Methods of Application

- Functional Resonance Analysis Method (FRAM)
- Process Hazard Analysis (PHA)
- Formal Hazard Assessments (JSAs, THAs, etc.)
- Proactive use of root cause analysis tools
  - TapRooT©, 5-Why, Lotus Blossom, Bow Tie, etc.

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HAZOPs

- Generally used as a very detailed and structured examination of individual components of a process or system
- Typically done as part of the design stage of a project, usually after preliminary drawings are completed
- Focus is usually on system operability issues rather than worker – system interfaces and OHS hazards
- Best practical time for a safety profession to get his / her perspective on the table
  - Rare to be part of project conceptualization

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HAZOPs

- For new facilities (greenfield) multiple HAZOPs on different systems would be ideal
- Should also push for consideration of impacts on workers and other systems (i.e., pedestrian vs. vehicles)
- Even at the design stage it can become difficult to get basic OHS considerations on the table
  - Budgets, nice-to-have elements, sunk cost thinking, process-centred thinking

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- Detailed Engineering drawings – very useful, need to be able to read
- HAZOP graphical example – software
- 3D representation of design for HAZOP

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Failure Modes & Effect Analysis (FMEA)

- Similar to a HAZOP, but often less detailed and may lack rigid structure in application
- Like HAZOPs, usually limited to very specific process and system
- Often doesn’t examine detailed blueprints – usually a “brainstorm”

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Cognitive Work Analysis (CWA) (Read)

"The CWA framework has five phases of analysis:

- Work domain analysis (WDA)
- Control task analysis (ConTA)
- Strategies analysis (SA)
- Social organisation and cooperation analysis (SOCA)
- Worker competencies analysis (WCA)."

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AIHS (Australian Institute of Health and Safety). (2019). Health and Safety in Design. In *The Core Body of Knowledge for Generalist OHS Professionals*. Tullamarine, VIC: Australian Institute of Health and Safety, p. 21.

PETE Analysis (Fiona Begg)

People: who is using the facility, who may be present (employees, visitors, contractors, suppliers, vendors, public, etc.)

Equipment: used by people, present at facility, maintenance considerations, lifecycle management, human-equipment interfaces

Tasks: what are people doing (production), maintenance considerations, human-task challenges, demands on persons

Environment: work environment and impacts, maintenance considerations, disposal and wastes

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AIHS (Australian Institute of Health and Safety). (2019). Health and Safety in Design. In *The Core Body of Knowledge for Generalist OHS Professionals*. Tullamarine, VIC: Australian Institute of Health and Safety, p. 21.

Misconception Analysis (Busby)

- Focuses on identifying areas of misconception / misinterpretation in human-system operations
- Questioning assumptions of:
  - Operator action by designers – operator adaptation, attention, and procedural adherence, etc.
  - Systems by operators – reliance on intuition, past experience, procedural rigidity, system resilience to errors, etc.

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AIHS (Australian Institute of Health and Safety). (2019). Health and Safety in Design. In *The Core Body of Knowledge for Generalist OHS Professionals*. Tullamarine, VIC: Australian Institute of Health and Safety, p. 21.

Functional Resonance Analysis Method (FRAM)

- Modelling of system function as it actually performs
- Main areas of use include:
  - Identifying potential for unintended events
  - Testing proposed systems and changes

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AIHS (Australian Institute of Health and Safety). (2019). Health and Safety in Design. In *The Core Body of Knowledge for Generalist OHS Professionals*. Tullamarine, VIC: Australian Institute of Health and Safety, p. 21.

Process Hazard Analysis (PHA)

- Main method of managing hazardous chemical processes in plant environments
- Similar to HAZOP, but used as ongoing safety management program for chemical process systems
- Limited in scope and does not include non-process demands (layout and spacing, other human-system interfaces)

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Formal Hazard Assessments

- Typically limited to a specific job / position or a defined task
- Usually the minimal legal requirement to meet OHS due diligence
- Safety by Design would require formal hazard assessments on all activities and operations to be completed, reviewed as a bundle, and connected elements to be assessed and addressed

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Root Cause Analysis Techniques

- TapRooT©
- 5-Whys
- Lotus Blossom

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Safety by Design Interventions

- Ensuring a new staircase is fitted with proper handrails on both sides
- Eliminating confined spaces from new structures
- Adding fixed ladders or catwalks so maintenance personnel can safely access equipment as necessary
- Placing bollards at common locations where pedestrians and vehicles are in close proximity

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Safety by Design Interventions

- Putting short stairs over tank containment walls to avoid having to climb over the walls
- Placing heating systems in walkway surfaces to keep them snow and ice free in winter
- Adding heavy bumpers around racking footings to prevent forklift hits to the structure
- Ensuring access platforms are of sufficient width and depth for a person to stand on them and perform required tasks

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Safety by Design Interventions

- Separating employee parking and entrances from heavy vehicle traffic
- Placing paint room in far corner to better install and maintain required ventilation system
- Have ventilation fans moved from cramped crawlspace to accessible service room
- Add stairs over insulated piping so that workers will not fall or damage insulation when taking common shortcut

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Safety & Procurement

- Every operation involves the purchase of items that impact the health and safety of workers
- Having specific safety standards in a procurement process helps to prevent the introduction of hazards into the workplace

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Limiting New Hazards

- Identifying safer chemicals and “banning” the purchase of other chemicals
- Mandating all portable power tools be double-insulated
- Limiting purchase of supplies to select suppliers and or product lists that meet organizational OHS requirements

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Safety in Procurement

- Requiring specific ergonomic office furniture, IT devices
- Allowing only higher-quality grinding disks to limit risk of shattering
- Mandating anti-fog safety glasses in all situations
- Requiring maximum noise thresholds for new machinery

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Management of Change

- Managing changes to operations – process or inputs – is another factor to consider
- Application of the formal hazard assessment process assists this in operational situations
- Having management of change requirements in procurement – which includes a safety review – is another element in a Safety by Design approach
- Don’t need to reinvent the wheel just for safety

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